PRISM | Tech Management, SRI-B

CLASH OF THE CLAWS

RV College of Engineering_AutoMind Collective
Your phone becomes a different device depending on where you are

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Problem Statement

The Core Challenge

Smartphones are powerful but **context-blind** — they behave identically in an exam hall, a gym, and a job interview.

The Manual Burden

Users manually toggle DND, hide apps, and change ringtones every context switch — often forgetting under stress.

Affected Users

Students, professionals, interview candidates, and developers switching environments daily.

Why It Matters

Privacy leaks, 20+ min productivity loss per switch, and zero proactive prep for critical events.

Supporting Evidence

- **Focus Drain:** Takes ~23 min to regain deep focus after a switch (UC Irvine).
- **Cost of Interruption:** Knowledge workers lose 2–3 hours daily (Microsoft).
- **Reactive Habits:** Avg. user checks phone 47x/day, mostly reactively (Deloitte).
- **Setup Fatigue:** 68% disable Focus Mode due to tedious manual setup (Samsung).

Current Solution & Gaps

Existing Solutions: Samsung Focus Mode, Apple Focus, Tasker, IFTTT, Google Digital Wellbeing

1. No Behavioural Learning

Requires manual rules. Doesn't observe patterns or build intelligent schedules. No smarter after a week.

2. Reactive, Not Proactive

Fires *after* context entry. Phantom Mode switches 15 mins ahead to ensure you're ready.

3. No Conflict Intelligence

Can't handle overlapping events. Doesn't surface conflicts or ask for user preference in real-time.

4. No Contextual Pre-loading

Controls settings only; doesn't prepare assets (resumes, notes) before critical events occur.

The Opportunity: An always-on agent that watches 7 days of behavior, builds your persona map automatically, and acts proactively using an LLM reasoning layer.

Our Solutions

What we built: Phantom Mode

An OpenClaw-inspired Android agent that creates behavioural personas for your phone and activates them before you need them.

How it works

Periodic context engine updates pi-engine with modular sensor/API signals ⟶ Pi Engine decides persona ⟶ ADB commands apply device actions in near real-time.

6 Personas

Work, Gym, Interview, Home, Social, Deep Work

Proactive Switching

Switches 15 min before predicted context, not after arrival.

Conflict Resolution

Via Telegram — choose between clashing personas with one tap.

Pre-loading Assets

Trigger app/content opening based on persona decision.

AI Persona Suggestions

Integrated dual-API fallback for persona generation from 7-day routines, ensuring uninterrupted pattern-based suggestions with user validation.

Demo/Product Walkthrough

Manual Override

```
🌐 API WRAPPER / MOCK SERVER IS RUNNING

📡 Port: 5000
🔗 GET    /health    — Health Dashboard
🔗 POST   /decision  — Trigger & fetch actions
🔗 POST   /context   — Send sensor data
🔗 POST   /override  — Send user input
🔗 POST   /ack       — M2 Action acknowledgement

📡 M4 live push: ENABLED -> http://127.0.0.1:8000

🛡️ Fallback Failsafe Mode: ALWAYS ACTIVE
```

```
PS C:\Coding\PrismOpenClaw\pi-engine> node apiWrapper.js
   battery        [▓▓▓▓▓▓▓▓▓▓] 98%  →  85
   activity       [▓▓▓▓▓▓▓▓▓▓] 80%  →  "working"
   time           [▓▓▓▓▓▓▓▓▓▓] 100% →  18

🔍 End of Decision Explanation

🔄 TRANSITION ENGINE: none → work
   Type: INSTANT
   Message: "🔄 Persona updated."
   ✅ Transition complete → work

⚙️EXECUTOR: Activating persona "work"
```

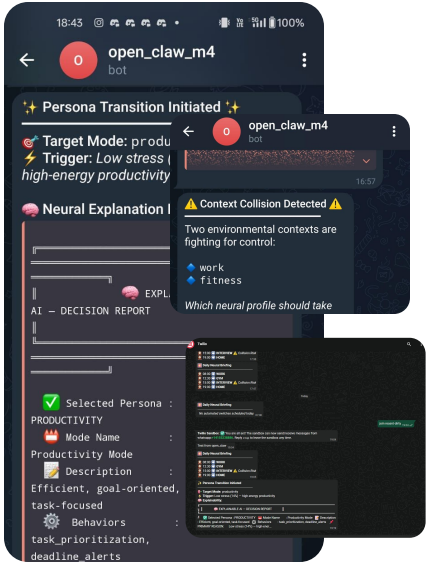
ADB Context Engine

```
PS C:\Coding\PrismOpenClaw\m2-android-adb> node index.js --url http://localhost:5000/decision/latest
[2026-05-08T12:18:05.050Z] INFO Loaded 24 actions from C:\Coding\PrismOpenClaw\m2-android-adb\actions.json
Connecting to ADB 127.0.0.1:5037...
[2026-05-08T12:18:05.065Z] INFO Executing action 'disable_dnd'
▶ Executing action 'disable_dnd'
[2026-05-08T12:18:05.216Z] INFO Successfully executed 'disable_dnd' in 150ms
✅ Executed 'disable_dnd' (150ms)
📤 Sending ack to M4: http://localhost:5000/ack
✅ Ack sent successfully
PS C:\Coding\PrismOpenClaw\m2-android-adb>
```

```
PS C:\Coding\PrismOpenClaw\pi-engine> node apiWrapper.js
   🔗 GET    /health    — Health Dashboard
   🔗 POST   /decision  — Trigger & fetch actions
   🔗 POST   /context   — Send sensor data
   🔗 POST   /override  — Send user input
   🔗 POST   /ack       — M2 Action acknowledgement

📡 M4 live push: ENABLED -> http://127.0.0.1:8000

🛡️ Fallback Failsafe Mode: ALWAYS ACTIVE

👍 [M2] Actions Acknowledged: disable_dnd
```

```
🧠 M3 CONTEXT ENGINE - PrismOpenClaw Phantom Mode
==========================================================
Intelligent Context Sensing & Proactive Prediction
==========================================================

📋 Loading configuration...
   M1 API URL: http://localhost:5000
   Heartbeat Interval: 60s
   Simulation Mode: True
   Demo Mode: True

🔧 Initializing system components...
```

Telegram Interaction (and WhatsApp via Twilio)



Tech & Architecture

Tech Stack

- **Decision Engine:** Node.js REST APIs for context, decisions, and health checks.
- **Android Executor:** Node.js CLI with ADB mappings for live execution.
- **Context Intelligence:** Python service for periodic signal generation.
- **Communication Layer:** FastAPI with Telegram/WhatsApp for user alerts.
- **Contract:** JSON over REST with token-authenticated endpoints.

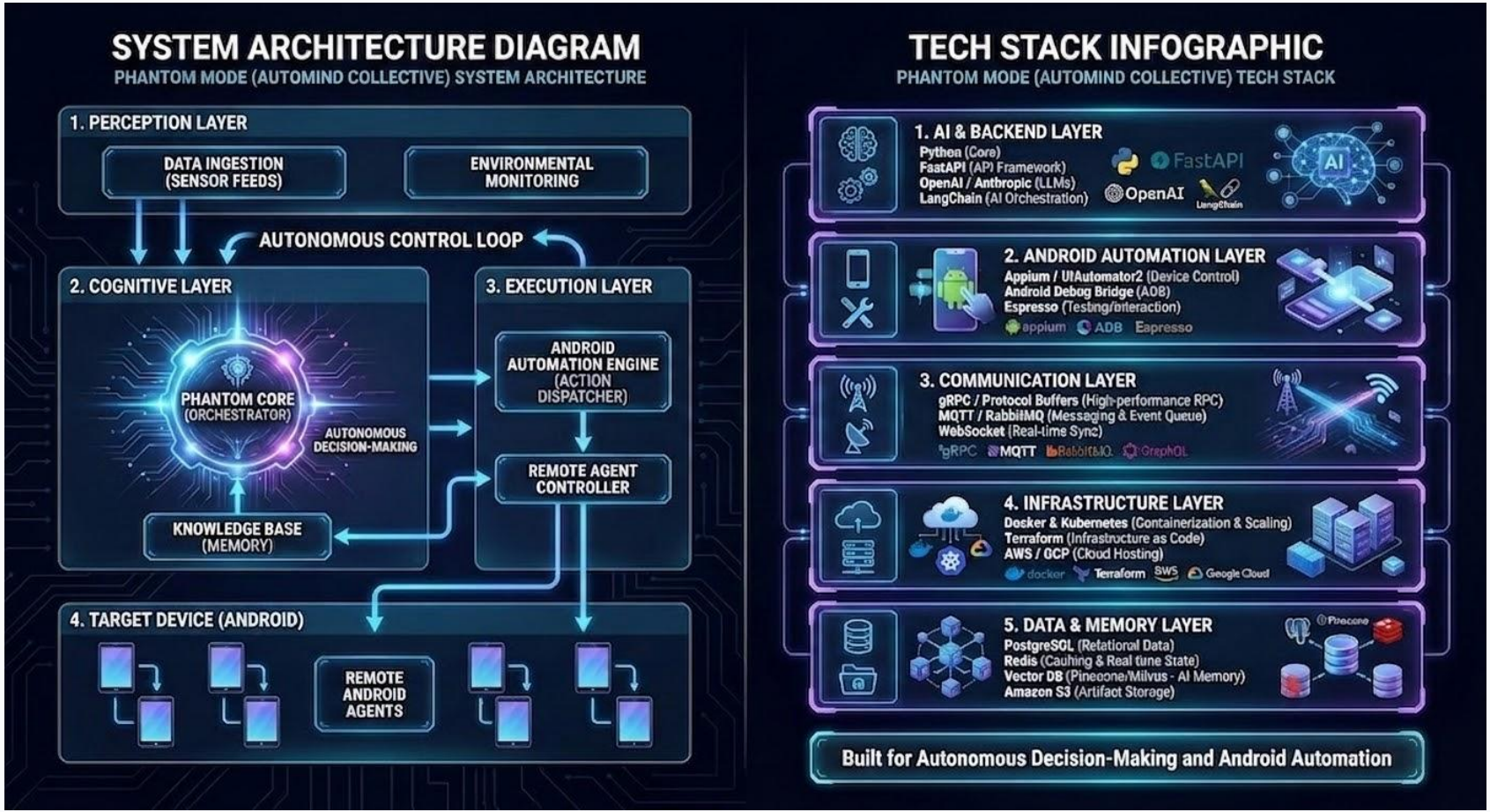
Architecture Flow

Context → Decision → Action/Messaging → Feedback Loop → Engine

Design Properties

- **Single Source of Truth:** Centralized decision engine.
- **Human-in-the-loop:** Control via Telegram buttons.
- **Safety Fallback:** Dry-run mode for restricted ADB.
- **Graceful Degradation:** Robust to integration failures.

Phantom Mode



Impact & Use cases

Student

Walks into exam hall → social apps locked, notes opened, silent mode on — zero manual action

Professional

Enters office GPS zone → corporate email foregrounded, personal chats silenced, neutral wallpaper set

Interview Candidate

15 min before interview → resume loaded, briefing sent to WhatsApp, formal ringtone set

Developer

Tuesday 2pm (routine deep work slot) → all notifications off, focus playlist starts, greyscale mode on

Scale & Integration

6.8B Android users · Samsung One UI Good Lock integration path · Enterprise MDM compliance angle

Differentiation

Why we stand out: Our prototype differentiates itself through proactive decisioning, explainability output, and one-tap human override in a closed feedback loop.

MOAT 1

Behaviour memory

7-day routine stored in `memory` grows more accurate every week.

A competitor starting fresh has zero personalised history — a data moat that deepens over time.

MOAT 2

Persistent reasoning

Runs a continuously reasoning like an OpenClaw agent that watches, learns, and acts without being triggered.

Unlike competitors' one-shot calls, this requires implementation like the full OpenClaw stack.

MOAT 3

Portable Persona Sharing

Human-readable markdown files that users can share, edit, and trade in a marketplace.

A behavioural theme store for your OS — an abstraction layer no one else has.

Access & Documentation

GitHub Repository

Access the complete source code and implementation details.

Hrithik-Bhattacharya/PrismOpenClaw

Live Demo

[Demo Link Placeholder]

AI Disclosure

Document submitted for transparency and compliance.

Samsung PRISM & Tech Depth

Samsung PRISM Alignment

- **On-device automation:** Applying Android behavior via ADB execution.
- **Context-aware:** Dedicated service feeds environment signals for proactive switching.
- **Human-in-control:** Instant overrides via Telegram conflict buttons.
- **Multi-channel UX:** Alerts via Telegram and WhatsApp surfaces.
- **Explainable AI:** Detailed reasons for every mode transition.

Technical Depth

- **Modular Architecture:** Separation of intelligence, orchestration, and execution.
- **Closed-loop Control:** Deterministic execution with feedback re-entry.
- **API Contracts:** Token-protected REST endpoints and JSON payloads.
- **Safe Design:** Dry-run modes and restricted-device fallback strategies.
- **Resilience:** Graceful degradation of optional third-party integrations.

"Phantom Mode is an OpenClaw-inspired, context-adaptive mobile automation architecture that demonstrates explainable AI decisions, human override, and safe execution paths suitable for Samsung ecosystem evolution."